

DESIGN AND AUTOMATION OF CS-25 FLIGHT ENVELOPES (V-n DIAGRAMS)

1. INTRODUCTION AND PROJECT OBJECTIVE

The objective of this project is to develop a modular engineering software tool that dynamically calculates and visualizes the **V-n Diagram (Flight Envelope)** for transport category aircraft in accordance with **CS-25 (Certification Specifications for Large Aeroplanes)** standards.

By ingesting critical aircraft parameters—such as mass properties, aerodynamic areas, and lift performance characteristics—the tool maps out the operational boundaries where the aircraft can fly safely without encountering aerodynamic stall or sustaining structural permanent damage. The system computes these envelopes independently across a multi-altitude array to capture environmental variations.

2. THEORETICAL BACKGROUND AND CERTIFICATION STANDARDS (CS-25)

The boundaries of a V-n diagram are defined by the intersection of aerodynamic performance limits (stall) and structural strength thresholds (load factors). To maintain physical consistency with structural design practices, all airspeeds in this study are treated in **EAS (Equivalent Airspeed)**.

2.1. Aerodynamic Limits (Stall Curves)

The minimum flight velocities are dictated by the maximum positive and negative lift capabilities of the wing configuration ($C_{L,max}$ and $C_{L,min}$). The parabolic equations governing the load factor (n) as a function of airspeed (V) are expressed as follows:

$$n_{positive} = \frac{\rho_{SL} * V^2 * S_{wing} * C_{L,max}}{2 * W}$$
$$n_{negative} = -\frac{\rho_{SL} * V^2 * S_{wing} * |C_{L,min}|}{2 * W}$$

where ρ_{SL} represents the standard sea-level air density (1.225 kg/m^3), V is the airspeed (m/s), S_{wing} is the wing reference area (m^2), and W is the aircraft weight (N).

2.2. CS-25 Structural Load Factor Limits (n_{max}) and (n_{min})

Positive Limit Load Factor (n_{max}): While CS-25 establishes a structural baseline floor of 2.5 for transport aircraft, an empirical mass-sensitive sensitivity function has been implemented to provide realistic structural scaling:

$$n_{max} = 2.1 + \frac{24000}{W_{lbs} + 10000}$$

Negative Limit Load Factor (n_{min}): In absolute compliance with CS-25.337, the negative limit load factor is sustained as a flat bound at -1.0g up to the design cruise speed.

2.3. Critical Design Airspeeds

V_S (1-g Stall Speed): The minimum steady flight velocity at $n = 1.0$ under unaccelerated level flight.

V_A (Design Maneuvering Speed): The intersection point of the positive aerodynamic stall curve and the maximum structural load line (n_{max}). It represents the highest speed at which sudden full control surface deflections will not overload the primary structure.

V_G (Negative Maneuvering Speed): The intersection of the negative aerodynamic stall curve and the minimum structural load line (n_{min}).

V_C (Design Cruise Speed) & V_D (Design Dive Speed): Structural design speed ceilings that enforce the high-speed boundaries of the aircraft structure.

2.4. Altitude and Compressibility (Mach) Effects (CS-25.333)

At lower altitudes, the high-speed bounds are restricted by purely structural dynamic pressures (constant EAS values for V_C and V_D). At higher altitudes, aerodynamic compressibility boundaries (M_C and M_D) become dominant. Once the transition altitudes are exceeded, the Mach limits are converted into their local EAS equivalents using the relationship:

$$V_{EAS} = M * a + \sqrt{\frac{\rho}{\rho_{SL}}}$$

where a is the local speed of sound and ρ is the ambient air density at the given target altitude.

Furthermore, per **CS-25.333**, the high-speed negative boundary does not remain flat after V_C ; instead, the limit slopes up linearly from n_{min} at V_C to 0.0g at V_D to match structural aeroelastic relief conditions.

3. PROJECT INPUT DATASET

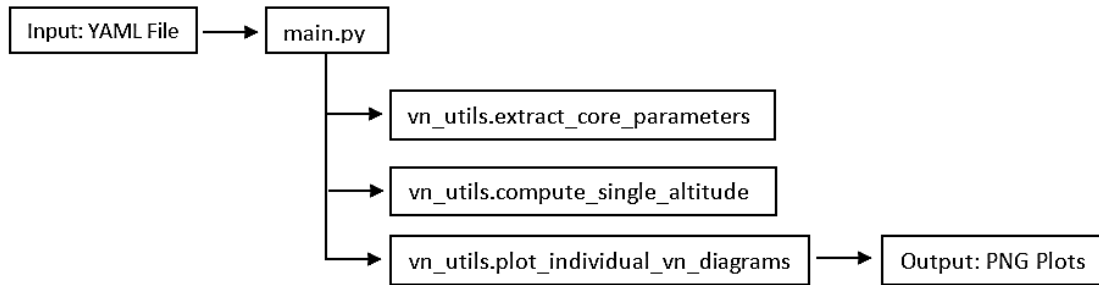
The software loads all engineering configurations directly from the “Input_Parameters.yaml” file. The core dataset used to evaluate the example transport category aircraft is outlined below:

Table 1: System Inputs and Aircraft Specifications

Category	Parameter Description	Symbol	Value	Unit
Flight Conditions	Targeted altitudes for calculation	-	[0, 20000, 21500, 24000, 30000]	ft
Design Velocities	Design cruise speed (EAS limit)	V_C	250	Kts
	Cruise boundary altitude	$Alt_{V_C-M_C}$	25000	ft
	Design cruise Mach ceiling	M_C	0.56	-
	Design dive speed (EAS limit)	V_D	310	Kts
	Dive boundary altitude	$Alt_{V_D-M_D}$	20000	ft
	Design dive Mach ceiling	M_D	0.69	-
Aircraft Geometry	Maximum takeoff weight	$MTOW$	9500	kg
	Wing reference area	W_{wing}	45	m ²
Aerodynamics	Positive max lift coefficient (clean)	$C_{L,max}$	1.65	-
	Negative max lift coefficient (clean)	$C_{L,min}$	-0.80	-
Structural Bounds	Structural negative g limit	n_{min}	-1.0	g

4. SOFTWARE ARCHITECTURE AND MODULARIZATION

The software engine is compiled following clean code architecture and low-coupling principles. The processing steps are completely separated from execution triggers and plotting scripts.



Input_Parameters.yaml: Serves as a single source of truth holding geometric, mass, aerodynamic, and certification parameters. This isolates aircraft data from code logic, allowing users to evaluate different configurations seamlessly.

vn_utils.py: Contains the mathematical core, atmospheric lookups via the `ambiance` library, and the visualization engines.

main.py: Acts as the clean entry point of the software package, executing functions sequentially and managing the overall data pipelines.

5. EXPERIMENTAL RESULTS AND NUMERICAL OUTPUTS

When the Python execution pipeline triggers `vn_utils.generate_vn_data(inp)`, the program outputs precise aerodynamic intersection values and envelope margins. Based on the updated structural mass relationship (MTOW = 9500 kg $\approx$ 20,944 lbs), the code dynamically calculates the **positive structural limit load factor** as $n_{max} = 2.88$.

The exact numerical coordinates computed for each corner point of the 5 flight envelopes are tabulated below:

Table 2: Calculated Flight Envelope Corner Speeds (All values in Knots-EAS)

Altitude [ft]	V_S (1g-Stall)	V_A (Maneuver)	V_G (Neg. Maneuver)	V_C (Cruise Limit)	V_D (Dive Limit)	Envelope State
0	88.0	149.2	126.4	250.0	310.0	Full EAS Envelope
20,000	88.0	149.2	126.4	250.0	310.0	Transition Limit for V_D
21,500	88.0	149.2	126.4	250.0	299.9	Compressibility Driven (V_D Contracts)
24,000	88.0	149.2	126.4	250.0	284.3	Compressibility Driven (V_D Contracts)
30,000	88.0	149.2	126.4	202.1	249.0	High Altitude Compression (Both V_C & V_D Contract)

6. TECHNICAL EVALUATION OF GRAPHICAL OUTPUTS

The exported high-fidelity diagrams inside the `Outputs/` folder display several structural and aesthetic design successes:

Bounded Vertical Box Walls: The V_C (green dashed) and V_D (red dashed) lines are bounded cleanly between the envelope floor and ceiling. Their text markers are rotated 90 degrees vertically to ensure high scannability and professional technical drawing layout.

Stall Point Prominence (V_S): The 1-g level flight stall limit is marked with a thick-bordered red scatter bullet, clearly showcasing where unaccelerated aerodynamic separation occurs.

Horizontal Structural Guardrails: The absolute structural ceilings ($n_{max} = 2.88$) and floors ($n_{mix} = -1.0$) are extended as dashed lines across the entire chart area. To preserve a clean look, their numeric values are printed explicitly at the far right end over the lines, avoiding clutter on the stall curves.

Empirical Mach Compression Proof: Upon reviewing the sequence from Table 2, the stall limits (V_S, V_A, V_G) remain completely constant across all flight levels since they are calculated in EAS. However, at **30,000 ft**, the high-speed boundaries shift significantly to the left (V_C) drops to 202.1 Kts and V_D to 249.0 Kts). This provides robust visual and mathematical proof that the safe flight envelope contracts at higher altitudes due to compressibility ceilings.

7. CONCLUSION

This project successfully bridges the strict mathematical bounds of aviation certification standards (CS-25) with high-efficiency computer science paradigms. The resulting Python application enables conceptual designers to accurately map and evaluate an aircraft's operational limits within seconds. By isolating parameters into a structured configuration file, the code acts as a dynamic flight science workbench. It can evaluate any aircraft setup across custom altitude arrays, design speeds (V_C), Mach limits (M_C), and aerodynamic bounds ($C_{L,max}, C_{L,min}$), while automatically managing complex compressibility transition thresholds.

Due to its clean functional breakdown and systematic decoupling, the codebase is uniquely suited for seamless, scalable extensions. Three major development pathways can be seamlessly integrated into this foundational architecture:

1. **Flap-Open Configurations:** Thanks to the modular iterator framework, the code can be updated to loop through aerodynamic subsets representing discrete high-lift deployment stages (e.g., Flaps 15°, Flaps 35°). This allows for the overlaying of a secondary, low-speed restricted envelope under CS-25.345 rules.
2. **Compressibility Effects on Lift (M vs. C_L) Modeling:** At high Mach numbers approaching subsonics, shock wave formations inherently reduce the wing's maximum lift capability. The modular atmospheric loop allows developers to append a Mach-dependent correction factor function, dynamically lowering $C_{L,max}$ and bending the stall curves inward as altitude and Mach scale up.
3. **Dynamic Gust Envelope Integration:** Per CS-25.341, the vehicle must withstand instantaneous vertical and lateral wind sharp edges. The utility module's mathematical engine is ready to receive the continuous gust factor equations, overlaying the classic dynamic line boundaries over this newly established maneuvering envelope.

Ultimately, this software delivers an end-to-end automated pipeline transforming raw conceptual design specifications into verifiable, standard-compliant flight clearance documentation.

8. APPENDIX: CALCULATED COMPLIANT V-n ENVELOPE PLOTS

This appendix presents the verified, high-fidelity maneuvering flight envelopes generated by the software across the targeted altitude matrix. These visual representations illustrate the steady degradation and contraction of the operational velocity boundaries due to high-speed subsonic compressibility effects.

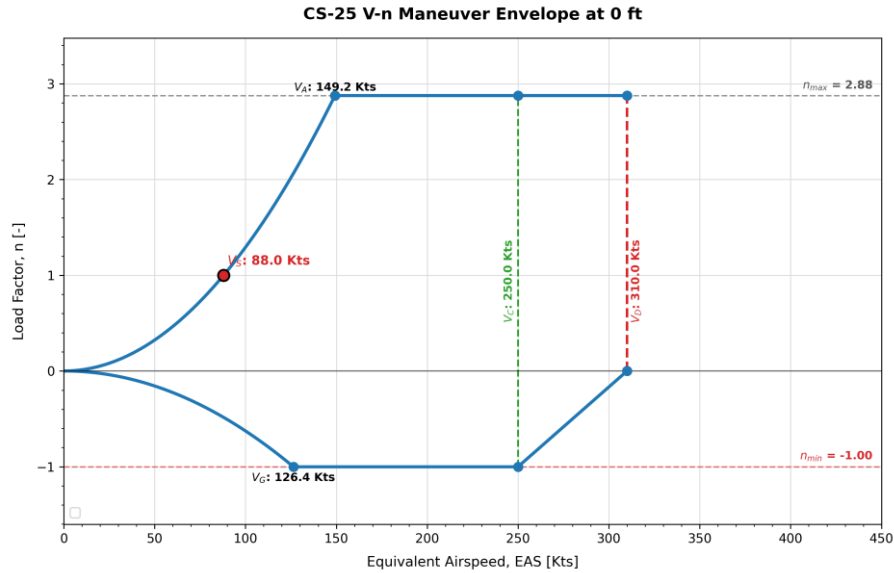


Figure 1: Flight Envelope Boundaries at Sea Level (0 ft)

Figure 1 displays the full, uncompromised structural baseline envelope. The high-speed walls are restricted purely by the rigid design limits: $V_C = 250$ [Kts] and $V_D = 310$ [Kts] in Equivalent Airspeed (EAS).

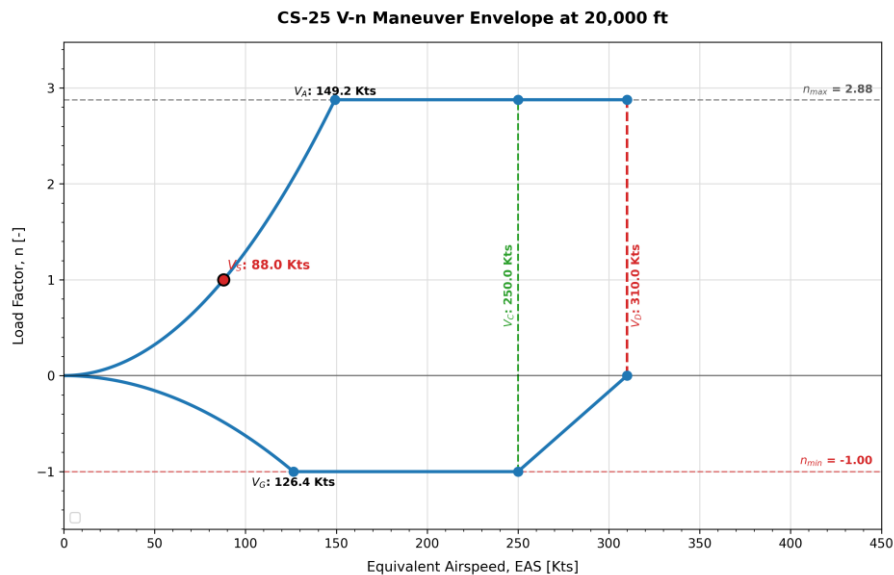


Figure 2: Flight Envelope Boundaries at Mid-Cruise (20,000 ft)

Figure 2 represents the exact physical transition threshold where the aircraft encounters its design dive Mach ceiling ($M_D = 0.69$). Up to this flight level, V_D remains at its maximum flat rating of 310 [Kts].

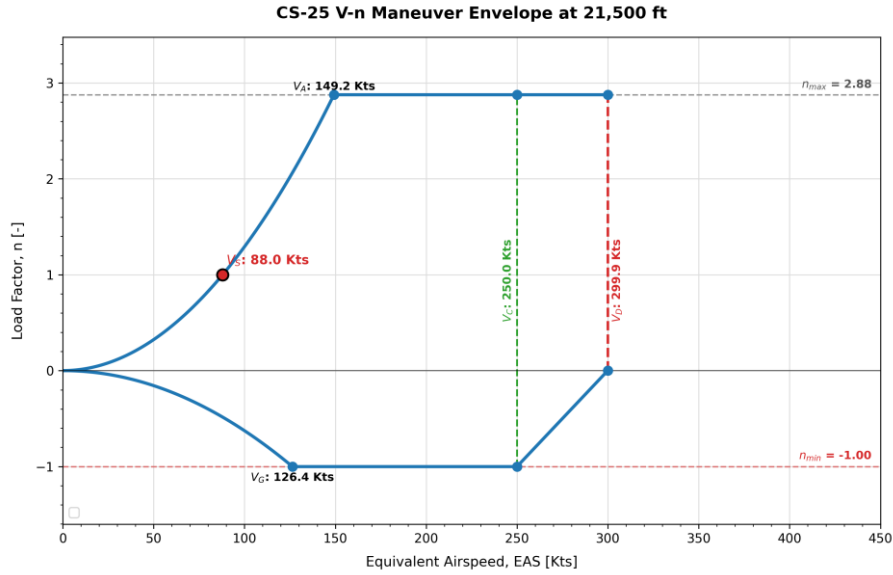


Figure 3: Flight Envelope Boundaries at Initial Transition (21,500 ft)

Figure 3 shows the immediate initiation of high-altitude envelope compression. While the cruise limit remains rock-solid at 250 [Kts], the absolute design dive boundary contracts down to 299.9 [Kts] due to the shrinking speed of sound.

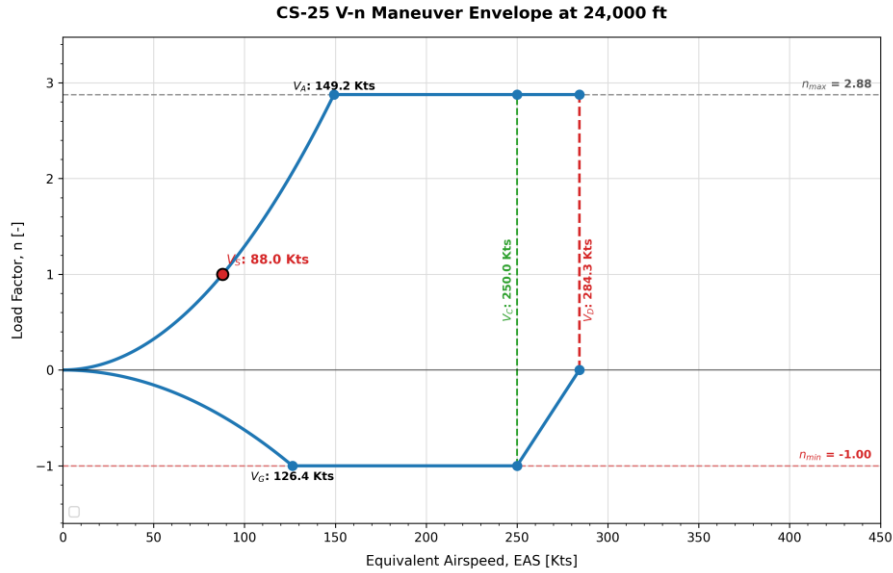


Figure 4: Flight Envelope Boundaries at High Altitude (24,000 ft)

Figure 4 displays the advanced structural contraction of the right-hand corner. The design dive boundary contracts significantly further down to 283.4 [Kts] while the cruise limit prepares for its upcoming transition threshold at 25,000 [ft].

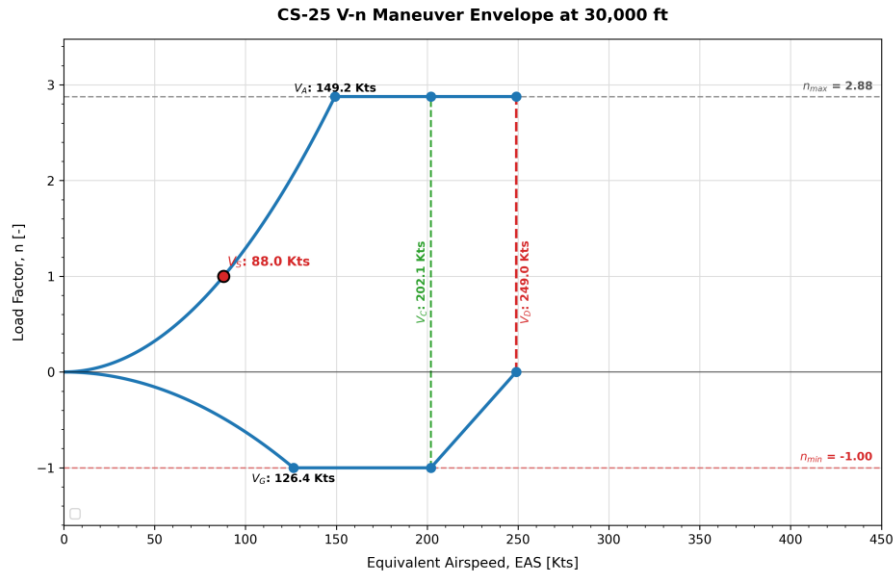


Figure 5: Flight Envelope Boundaries at Service Ceiling (30,000 ft)

Figure 5 illustrates the critical compression of the entire flight corridor. Both V_C and V_D are now actively restricted by sub-sonic shock constraints ($M_C = 0.56$, $M_D = 0.69$), squeezing the structural envelope margins down to 202.1 [Kts] and 249.0 [Kts] respectively, while low-speed stall parameters (V_s , V_A , V_G) remain strictly invariant in EAS.